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SHRISTI DAS BISWAS

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Education

Purdue University

Ph.D. in Computer Engineering; Advisor: *Dr. Kaushik Roy*

2021 – Present

GPA: 3.83/4.0

Indian Institute of Engineering, Science and Technology

B.Tech. in Electronics and Telecommunication Engineering

2017 – 2021

GPA: 9.39 /10.0; (Department Rank-2/61)

Technical Skills

Languages: Python, Git, Bash, MATLAB. **Tools:** Cadence Virtuoso, LTSpice. **Frameworks:** Pytorch, Tensorflow, L^AT_EX.

Internships

- Applied Science Intern at **Amazon Kumo, AWS** (Internship Manager: *Radhika Bhargava, Yue Zhang*) 2025 – 2025
- Applied Science Intern at **Amazon Fashion** (Internship Manager: *Prateek Singhal, Matthew Shreve*) 2024 – 2024

Publications

- S. Das Biswas, Y. Zhang, A. Pal, R. Bhargava, K. Roy. ELLA: Efficient Lifelong Learning for Adapters in Large Language Models. Accepted at **NeurIPS 2025 CCFM Workshop**, Under Review at **ARR**. [Link Awaiting Scipub Ticket Approval]
- S. Das Biswas, A. Roy, K. Roy. Now You See It, Now You Don't - Instant Concept Erasure for Safe Text-to-Image and Video Generation. Under Review at **CVPR 2026**. [Paper]
- S. Das Biswas*, A. Roy*, K. Roy. SlimDiff: Training-Free, Activation-Guided Hands-free Slimming of Diffusion Models. Under Review at **ICLR 2026**. [Paper]
- S. Das Biswas*, A. Roy*, K. Roy. CURE: Concept Unlearning via Orthogonal Representation Editing in Diffusion Models. Accepted at **NeurIPS 2025 (Spotlight)**. [Paper]
- S. Das Biswas, M. Shreve, X. Li, P. Singhal, K. Roy. PIXELS: Progressive Image Xemplar-based Editing with Latent Surgery. Accepted at **AAAI 2025**. [Paper]
- S. Das Biswas, E. Soufleri, A. Roy, K. Roy. Learning Unified Spatio-temporal Representations for Efficient Compressed Video Understanding. Accepted at **WACV 2026**. [Paper]
- A. Roy, M. Apolinario, S. Das Biswas, K. Roy. Feedback Alignment Meets Low-Rank Manifolds: A Structured Recipe for Local Learning. Accepted at **WACV 2026 (Highlight)**. [Paper]
- S. Das Biswas, A. Kosta, K. Roy. HALSIE - Hybrid Approach to Learning Segmentation by Simultaneously Exploiting Image and Event Modalities. Accepted at **WACV 2024 (Oral)**, **ICCVW 2023**. [Paper]
- S. Das Biswas, R. Chakraborty, A. Pramanik. A Brief Survey on Various Prediction Models for Detection of ADHD from Brain MRI Images. Accepted at International Conference on Distributed Computing and Networking **ICDCN 2020**. [Paper]
- S. Das Biswas, R. Chakraborty, A. Pramanik. On Prediction Models for the Detection of Autism Spectrum Disorder. Accepted at International Conference on Computational Intelligence in Pattern Recognition **CIPR 2020**. [Paper]

Ongoing Project

Developing training-free, efficient finetuning datasets for **Large Vision-Language Models** to select small, diverse, alignment-preserving subsets of image-text data for in-context learning. The goal is to enable efficient, globally-aware demonstration retrieval that improves multimodal alignment and reduces hallucinations with significantly lower cost.

Selected Experiences

Amazon Kumo, AWS

Bellevue, WA

Developed *Continual Learning Algorithms for Large Language Models*

2025 – 2025

- Researched and developed continual learning strategies for LLMs using **parameter-efficient fine-tuning (e.g., LoRA)** to support **scalable lifelong adaptation** using novel regularization techniques to mitigate task interference and catastrophic forgetting during sequential task learning. Provided a **formal theoretical analysis** of ELLA's properties.
- Evaluated model performance across multiple benchmarks (e.g., **DBPedia, Amazon Reviews, GLUE, SuperGLUE, GSM8k, BBH** etc.) using metrics like Backward Transfer (BWT), Forward Transfer (FWT), and task-wise accuracy, with **relative accuracy gains of up to 9.6% and a 35× smaller memory footprint across LLaMA and T5 models**.
- Benchmarked against **Claude Haiku 3.5** and **Claude Sonnet 3.5 v2** to achieve **upto ~59× lower latency** and up to 99% **lower annual costs** for improvements **upto +10% Precision and +12% Recall** compared to the AWS internal tool.

C-BRIC Lab, Purdue University

West Lafayette, IN

Developed a *Fast and Interpretable Training-Free Concept Unlearning Framework for Diffusion Models*

2025 – 2025

- Presented CURE, a **strong, scalable and training-free concept unlearning method** leveraging **orthogonal projections and spectral geometry**, to yield a closed-form weight update operator, dubbed **Spectral Eraser**, for **reliable and responsible** visual content creation in **T2I** models.

- Extensive experiments show **effective and robust removal of unsafe content, artist-specific styles, object and identities by up to 16%**. Significantly outperforms existing training-based and training-free methods in terms of generation quality, efficiency, specificity, and resistance to adversarial red-teaming tools for both **single- and multi-concept removals**.

C-BRIC Lab, Purdue University

West Lafayette, IN

SlimDiff: Training-Free, Activation-Guided Hands-free Slimming of Diffusion Models

2024 – 2025

- Proposed the **first closed-form, activation-guided structural compression of Stable Diffusion Models** that is entirely **training-free and timestep-aware** to guide dynamic model pruning under a user-specified compression budget.
- Evaluated SlimDiff on **MS-COCO, LAION Aesthetics, ImageReward, PartiPrompts** and **human-preference scoring using HPS v2.1 and Pic-a-Pic v1**. Achieved up to **35% acceleration** and **~100M parameter reduction over baselines** with no degradation in generation quality and formal theoretical guarantees of bounded pruning error.

Fashion and Fitness Org, Amazon Fashion

Sunnyvale, CA

Designed PIXELS, an Approach for Progressive Exemplar-Driven Image Editing in Diffusion Models

2024 – 2024

- Proposed an **off-the-shelf image editing tool using diffusion models** to enable customization by providing granular control over edits with reference images, allowing adjustments at the pixel or region level. Our method **operates solely during inference** for imitative editing, enabling users to draw inspiration from a **dynamic number of reference images** and **multi-modal prompts** to progressively incorporate all the desired edits without retraining or fine-tuning existing generation models.
- Demonstrated that PIXELS delivers high-quality edits efficiently, **outperforming existing methods in both exemplar-fidelity and visual realism by over 41% and 33.6% respectively** through quantitative comparisons and a user study.

C-BRIC Lab, Purdue University

West Lafayette, IN

Designed a Unified Representation Learning Approach for Compressed Video Understanding

2023 – 2024

- Proposed a lightweight yet powerful factorized end-to-end framework for **compressed video understanding** by revamping dominant raw video architectures since we find **decompressing videos is not only an overhead but also an inconvenience**. Our design made representations **more robust and reduced dimensionality**, making training less computationally challenging.
- Achieved SOTA video recognition performance on UCF-101, HMDB-51, K-400, and SS-v2 datasets with **favorable costs (330× higher)** and **fast inference (15× higher)**, inspiring future work toward efficient video systems not requiring decoded videos.

C-BRIC Lab, Purdue University

West Lafayette, IN

Designed RSFormer, an approach to Recurring the Spike Transformer for Object Tracking

2023 – 2023

- Proposed a **multi-scale feature extraction backbone** to generate compact feature representation from event frames for downstream processing. Designed a **hierarchical transformer encoder with recurrent networks of spiking attention blocks** to obtain temporally-guided coarse and fine features fused using a lightweight MLP decoder for prediction.
- Achieved comparable performance to SOTA on event datasets 1Mpx and Gen1 with **up to 20× higher** parameter efficiency.

C-BRIC Lab, Purdue University

West Lafayette, IN

Designed a Low-power Hybrid Approach to Learning Scene Segmentation using Event-Vision

2021 – 2022

- Proposed HALSIE, a novel **SNN+ANN hybrid** spatio-temporal feature integration approach to learning segmentation by simultaneously leveraging image and event camera modalities, enabling **efficient multi-modal learning**.
- Outperformed SOTA semantic segmentation benchmarks on DDD-17, MVSEC and DSEC-Semantic datasets with **up to 33.23× higher** parameter efficiency and **20× lower** inference cost, suitable for resource-constrained edge applications.

Indian Institute of Engineering, Science and Technology (Project Internship)

Shibpur, India

Designed Automated Predictive Models for ADHD and ASD Diagnosis

2019-2020

- Proposed a deep convolutional neural network model developed on **Tensorflow** framework to analyse and classify large **resting state fMRI datasets** for **ADHD and ASD diagnosis**.
- Achieved **up to 6% improvement** in accuracy on SOTA benchmarks within only 30 epochs of training. Our work has been communicated and is under review at the Journal of Neuroscience Methods, Elsevier

Reviewer Service

NeurIPS, AAAI (**Program Committee**), CVPR, ICLR, ACL, ICCV, WACV, IJCNN.

Relevant Coursework

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|-------------------------|-----------------------|--------------------------|------------------------|
| • Deep Learning (DL) | • Optimization for DL | • Reinforcement Learning | • Linear Algebra |
| • Data Structure & Algo | • Data Mining | • AI Hardware | • Advanced VLSI Design |

Projects

- * Deep spoken keyword spotting system.
- * Partitioning and layer assignment algorithm for TSV-aware 3D-IC structural planning.
- * Gesture controlled virtual mouse with canny edge detection.

Achievements

- * **NeurIPS 2025 Scholar Award.**
- * **IEEE Women in Engineering Best Student/ Research Scholar Paper Award:** Presented at ICDCN 2020.
- * **Google CS Research Mentorship Program Scholar.** Recipient class of 2023b.
- * **IEEE Women in Engineering Best Student/ Research Scholar Paper Award:** Presented at ICDCN 2020.
- * **Indian Academy of Sciences Summer Research Fellowship Program 2020.**
- * **Indian Youth Delegate to the People's Republic of China, 2019.** Organised by the Ministry of Youth Affairs and Sports, Government of India, and the Government of People's Republic of China.